Evaluating Chaotic Systems

Quantitative Evaluation of the Lorenz System

Omar Gamal ElKady

Nile University

July 2026

1 Part A — Phase Space Reconstruction

The optimal time delay was estimated from the scalar series $x(t)$ with `estimate_delay_ami` (first local minimum of the average mutual information), and the optimal embedding dimension with `estimate_dimension_fnn` (first dimension below the 1% false-nearest-neighbour threshold).

Setting	Value
AMIConfig	max_lag=200, n_bins=32, criterion=first_local_min
FNNConfig	max_dim=15, R_tol=10, A_tol=2, threshold_percent=1.0%, theiler=50
Result	$\tau = 16$ samples, $m = 3$

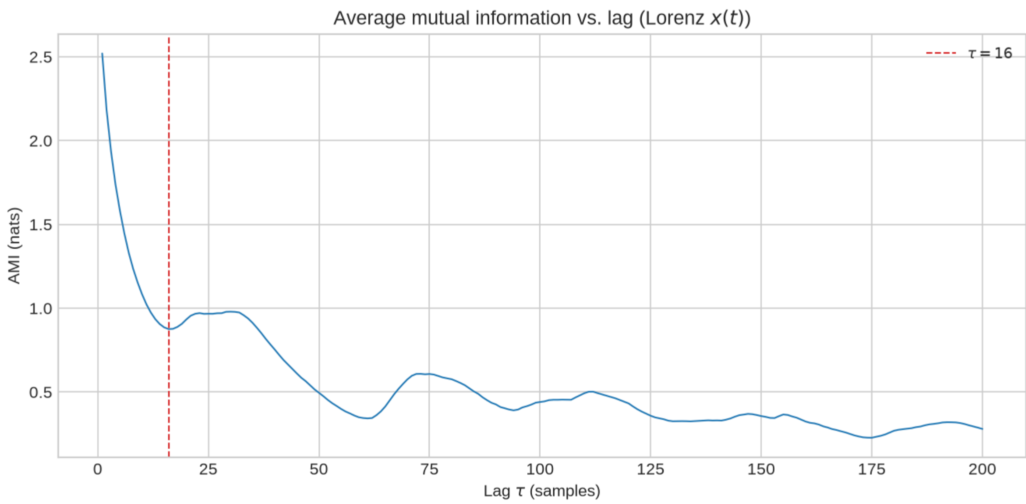

Figure 1: Average mutual information vs. lag, with the selected delay marked.

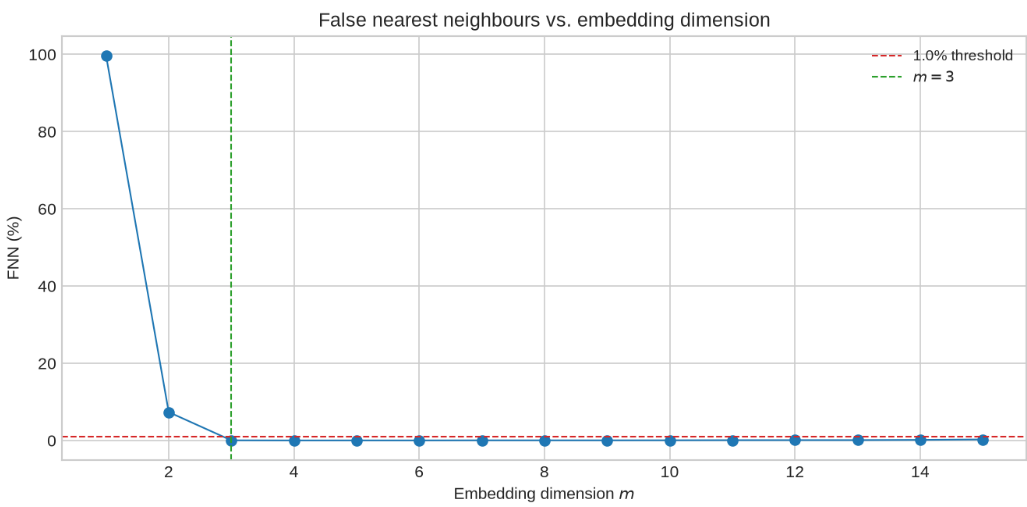

Figure 2: False-nearest-neighbour percentage vs. embedding dimension, with the 1% threshold and selected m marked.

The estimated delay ($\tau = 16$) and embedding dimension ($m = 3$) both fall inside the range expected for the Lorenz system ($\tau \sim 10$–17, $m \sim 3$–7), and m matches the true state dimension of the Lorenz equations, confirming that a faithful reconstruction was obtained from a single observable.

2 Part B — Lyapunov Exponent

The full spectrum was computed directly from the Lorenz equations and Jacobian using Wolf's Gram–Schmidt reorthogonalisation method, and the largest exponent was separately estimated from the scalar series with Wolf's phase-space tracking method using the τ and m from Part A.

Setting	Value
ODE method	`AttractorODEConfig`(dt=0.01, transient_steps=1000, n_steps=29000, manual RK4); `WolfODEConfig`(ortho_steps=20, log_base='e')
Time-series method	`WolfConfig`(max_replacements=300); $\tau = 16$, $m = 3$

Setting	Value
Lyapunov spectrum (ODE, nats/time)	$(0.9009,\ -0.0031,\ -14.5644)$
λ_1 from ODE spectrum	0.9009 nats/time = 1.2997 bits/time
λ_1 from scalar series (`wolf_mle`)	1.3591 nats/time = 1.9607 bits/time
Tracked segments	300 over 149.4 time units

The spectrum has exactly one positive exponent, one exponent close to zero (the flow direction), and one strongly negative exponent, confirming chaotic — not hyperchaotic — behaviour, consistent with Task 1. The time-series estimate (1.96 bits/time) is noticeably larger than the ODE-based ground truth (1.30 bits/time). Repeating `wolf_mle` at a larger embedding ($m = 6$, τ unchanged) brings the estimate markedly closer to the ODE value, so the discrepancy is attributable mainly to using the minimal FNN embedding ($m = 3$): it is large enough to remove almost all false neighbours on average but still leaves some residual folding that inflates the measured divergence rate. Finite series length and the tracking parameters (`evolve_cap`, `max_replacements`) are secondary contributors, as noted in the assignment.

3 Part C — Kolmogorov (K2) Entropy

K_2 was estimated from consecutive-dimension correlation sums, $K_2 \approx \frac{1}{\tau}\left\langle \ln \frac{C_m(r)}{C_{m+1}(r)} \right\rangle_r$, over the same scaling region used for the correlation dimension in Part D.

Setting	Value
K2 estimate (nats/time)	m=3→4: 3.52, m=4→5: 2.88, m=5→6: 2.57, m=6→7: 2.28, m=7→8: 1.96

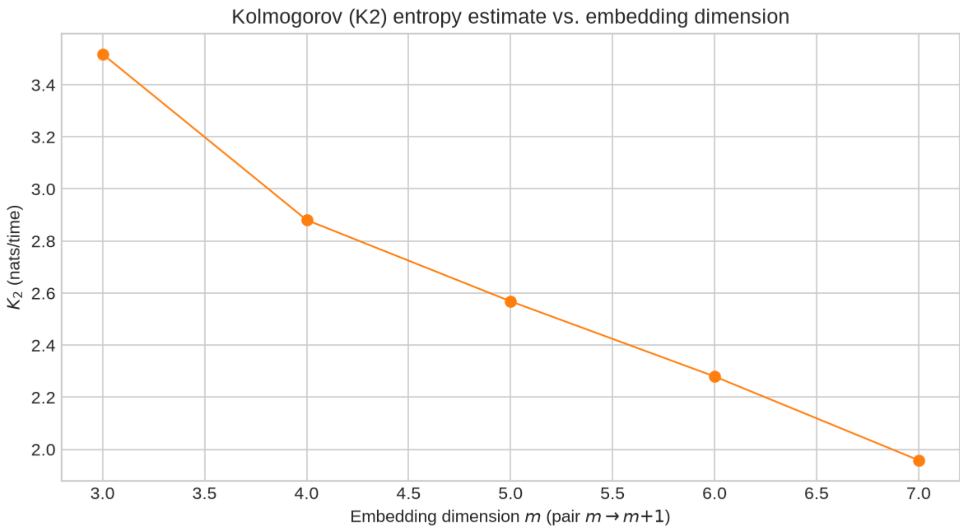

Figure 3: Kolmogorov (K2) entropy estimate vs. embedding dimension.

The K_2 estimate decreases monotonically as the embedding dimension increases, from 3.52 to 1.96 nats/time over the tested range, rather than growing without bound. This approach toward a finite value is the expected signature of low-dimensional deterministic chaos rather than noise, although the curve had not fully flattened by the largest embedding tested here; extending to a few more dimensions would be needed to pin down the exact saturated value.

4 Part D — Correlation Dimension

The Grassberger–Procaccia correlation sum $C(r)$ was computed with a Theiler window of $\tau \cdot m$ samples to exclude temporally correlated pairs, for embedding dimensions $m = 3$–8 at the τ from Part A. The correlation dimension D_2 is the slope of $\log C(r)$ vs. $\log r$ in the scaling region (marked in black).

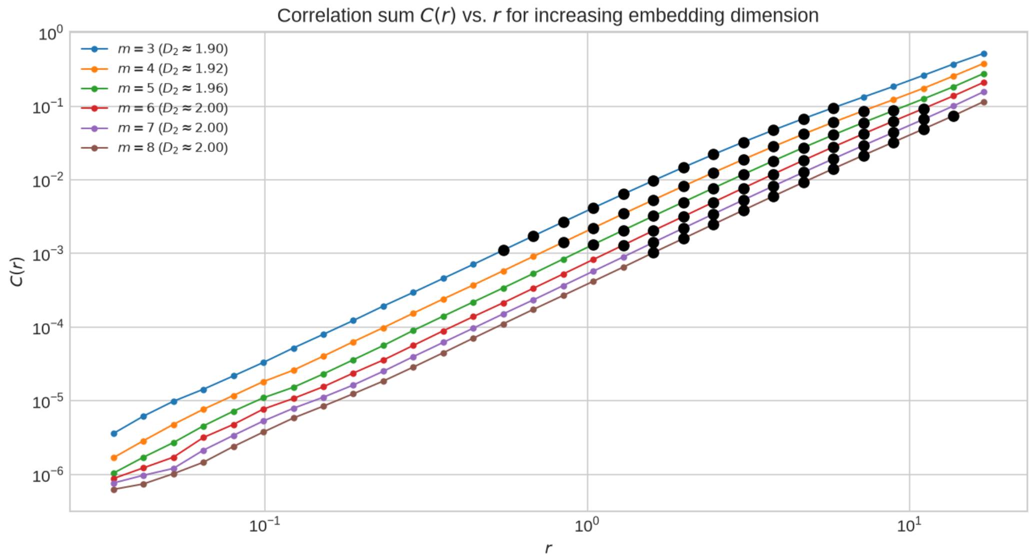

Figure 4: $\log C(r)$ vs. $\log r$ for increasing embedding dimension, with the fitted scaling region marked.

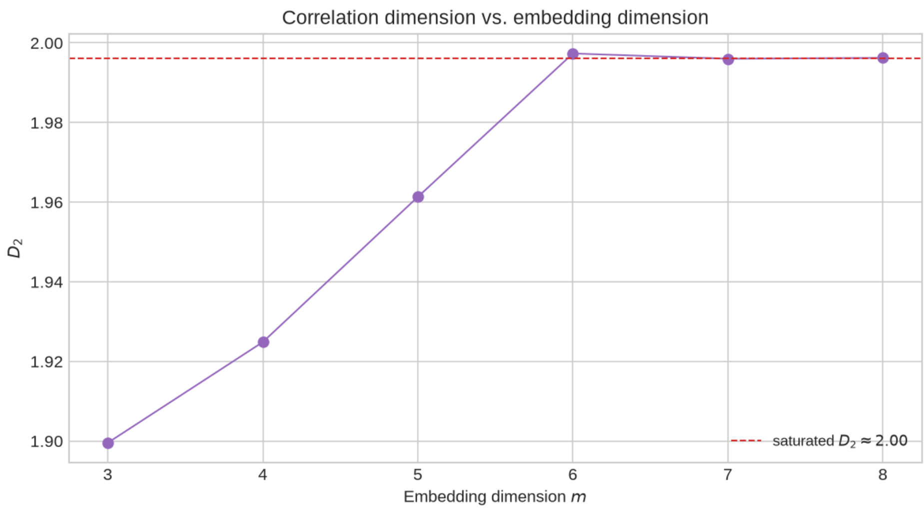

Figure 5: Correlation dimension D_2 vs. embedding dimension m.

D_2 rises from 1.900 at $m = 3$ to a saturated value of 1.996 by $m = 7$–8, in close agreement with the accepted correlation dimension of the Lorenz attractor (≈ 2.05). The clear saturation to a non-integer value confirms a low-dimensional fractal attractor.

5 Part E — Fractal (Box-Counting) Dimension

Box counting was applied directly to the original three-dimensional Lorenz trajectory using logarithmically spaced box sizes; D_0 is the slope of $\ln N(\varepsilon)$ vs. $\ln(1/\varepsilon)$ over the marked scaling region.

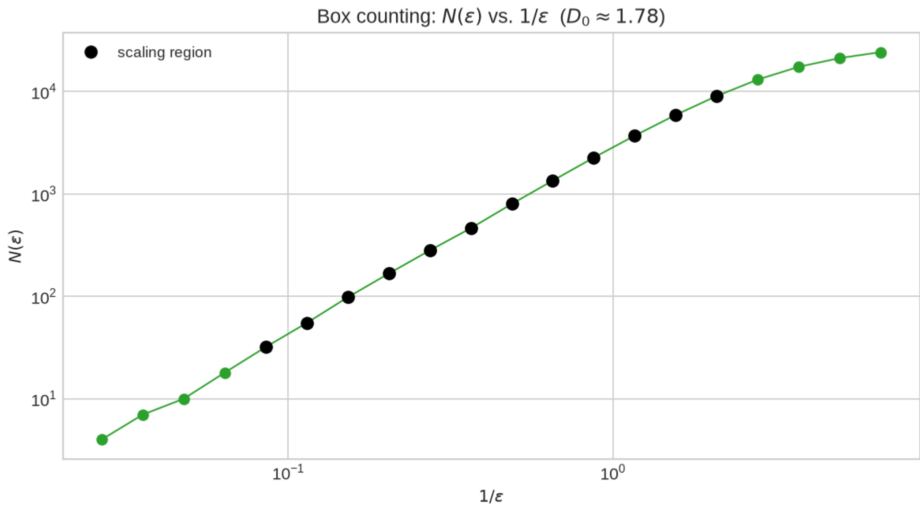

Figure 6: Box counting: $N(\varepsilon)$ vs. $1/\varepsilon$, with the fitted scaling region marked.

The box-counting estimate, $D_0 \approx 1.78$, is somewhat lower than the saturated correlation dimension ($D_2 \approx 2.00$), whereas $D_0 \geq D_2$ is generally expected for the same attractor. Box counting is known to need substantially more data than the correlation-sum method to resolve the same scaling range reliably, since it wastes resolution on empty boxes; the finite sample size used here is the most likely cause of the underestimate rather than a genuine violation of the $D_0 \geq D_2$ inequality.

6 Conclusion

The Lorenz attractor's Lyapunov spectrum (one positive, one near-zero, one strongly negative exponent), its positive and approximately saturating K_2 entropy, and its mutually consistent, non-integer D_2 and D_0 estimates are all independent signatures of the same underlying object: a low-dimensional, deterministically chaotic attractor rather than a noisy or hyperchaotic one. The reconstruction parameters recovered in Part A (τ and m) proved adequate to support every subsequent quantitative estimate, underscoring how much of this analysis rests on a correctly chosen embedding.